



CSE471: System Analysis and Design
Assignment on Functional Requirements

Proposed Project Title: FixNest - An On-Demand Multiple Home Services and Repair Platform

Group No: 02, CSE471 Lab Section: 08, Summer'2026		
SL	ID	Name
1	22201577	Sadia Binte Kamal
2	22201354	Sayeba Nasir
3	22201621	Rifah Nanjeeba Maruf

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Project Overview

The Home Services Platform is a comprehensive web application designed to connect homeowners with verified service providers such as electricians, carpenters, and plumbers for on-demand home repairs and maintenance, functioning for household services. The system manages three distinct user roles: Homeowners, Service Providers, and Admins. It facilitates location-based search and booking, secure job request management, and premium features such as priority scheduling and loyalty rewards, all supported by integrated payment, notification, and mapping systems.

Tech Stack:

- Language: React.js
- Framework: Node.js, Express.js
- Styling: TailwindCSS
- Database: MongoDB
- ORM: Mongoose
- Deployment: Vercel
- External APIs:
 - **Gmail API:** For sending booking confirmations, acceptance/rejection alerts, completion reminders, and other transactional emails. (Sadia)
 - **SSLCommerz API:** For processing service payments and premium subscription payments. (Rifah)
 - **OpenStreetMap API:** For location-based search and provider area mapping. (sayeba)

User Roles

1. **Homeowner:** The primary users. They can register, browse and search for service providers, request services, track job status, communicate with providers, and make payments.
2. **Service Provider:** A verified professional. They can manage their profile, set availability and pricing, accept or reject job requests, and track earnings.
3. **Admin:** A system super-user who is responsible for verifying service provider applications, monitoring platform activity, managing service categories and pricing, and resolving disputes.

Functional Requirements

SL	Common Workflows
1	Registration and Verification System: Homeowners can register by providing their name, email, phone number, password and address/location. Service providers go through a multi-step verification process that includes submitting licenses, documents, proof of background check, and area of expertise before becoming eligible to receive bookings.
2	Admin Dashboard & Access Management: An authorized administrator reviews pending service provider applications, checks submitted documents, and approves or rejects them. Approved providers are then visible in Home and Search Results and become bookable. Admins also monitor overall platform activity and manage service categories and pricing according to the service category.

Module 1	
Member	Feature Description
1	Service Provider Profile Management: Verified providers maintain a profile containing their qualifications, certifications, and experience on the services they are willing to offer. Providers add their service areas (neighborhoods/zones), set hourly or per job rates, and manage their daily availability and time slots so homeowners can only book open times. (sadia)
2	Search & Filter System: Homeowners can search for providers by service type (plumbing, electrical, carpentry, etc.) and filter results by location using Map, rating, price. They can view a provider's profile, reviews, and portfolio, then request a service by choosing a preferred date and time from the available slots. (Sayeba)

3	Request Management: Homeowners create service requests describing the problem and can track the request status (pending, accepted, in-progress, completed). Service providers can accept or reject incoming requests. (Rifah)
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Module 2	
Member	Feature Description
1	Service Provider Analytics: Each provider gets a private dashboard to track performance over time, including completed jobs, earned revenue, and rating trends. Providers can view monthly and yearly earnings reports, customer satisfaction metrics, and peak demand analysis by service type to identify their busiest periods. The average rating is found by dividing the total stars received by the number of reviews. (Sayeba)
2	Premium Facilities: Premium homeowners receive priority booking, allowing them to schedule services up to 2 days in advance useful, for situations such as needing an emergency electrician on a weekend. Premium homeowners receive an auto flat 5-10% discount on every booking. Premium users are exempt from the 24-hour cancellation fee, up to a limited number of times per month.(Rifah)
3	Favorite Providers List (Wishlist) and admin access: Homeowners can save providers to a "Favorites" list on their dashboard, so they can quickly book the electrician or plumber without searching again and the admin can see all the users and can remove their profile with a delete note. (Sadia)
4	Notifications & Email Reminders: Automatic mails keep both parties updated throughout the job lifecycle, including request confirmations sent to both homeowner and provider, acceptance/rejection alerts, completion reminders, and final job completion notifications. Premium users receive reminders and tracking for service warranties via email. (sadia)

Module 3	
Member	Feature Description
1	Request Management System: Homeowners may cancel a request, with a penalty applied if the cancellation occurs within 24 hours of the scheduled time, and either party can initiate a reschedule to keep the job on track which will be notified to the service providers through mail and a Messaging feature allows direct communication between the homeowner and provider throughout the job. (Rifah)
2	Review and Rating System: After a job is marked Completed, both parties can rate each other ; homeowners rate the provider's professionalism, quality, and punctuality, while providers rate the homeowner's behavior and payment promptness. Each profile displays an average rating calculated from all received feedback.(Sayeba) - MODULE 2
3	Subscription & Payment System: Users choose between Free and Premium plans. Free users get basic bookings at standard rates, while Premium costs 299 Taka per month or 2,990 Taka per year. Payments are processed securely through bKash/Nagad/Rocket etc. either at service completion or in advance, with an invoice generated for each service.(Rifah)
4	Off-Peak Pricing Heatmap Using existing analytics data (job counts by time), the system identifies low-demand hours and automatically offers homeowners a small discount for booking providers during those slots ("Book between 1-3 PM and save 10%"), helping providers fill idle time. Users get a simple heatmap-style dashboard showing demand by zone and hour. (Sadia)
5	Reward Points System: Users earn points per booking (for example, 5 points = 15 Taka) which can be redeemed for discounts on their next service, rewarding consistent platform usage. Premium users get to earn 100 taka more for renewing their membership. This feature rewards consistent users and encourages continued engagement with the platform. (Sayeba)

